

ROUYN-NORANDA AP, QC, Canada

WMO: 717255

Lat: 48.206N

Lon: 78.836W

Elev: 988

StdP: 14.18

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.2	-21.4	-35.6	0.8	-27.0	-29.9	1.1	-21.0	20.5	16.7	18.7	15.3	2.7	240	0.655

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	84.1	67.9	80.9	65.8	77.9	64.2	70.6	79.8	68.6	76.8	66.7	74.4	11.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.5	104.9	74.5	65.6	98.0	73.0	63.7	91.7	71.1	35.1	80.4	33.4	76.8	31.9	74.6	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.4	17.3	15.4		DB	-35.8	89.7	5.3	3.4	-39.6	92.1	-42.7	94.1	-45.6	96.0	-49.5	98.4
				WB	-36.0	74.3	5.2	2.0	-39.8	75.7	-42.8	76.9	-45.8	78.0	-49.5	79.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.4	3.5	7.1	19.3	34.2	49.7	59.7	63.9	61.4	53.9	41.5	27.3	13.0	
	DBStd	23.51	14.75	13.18	13.39	10.27	9.38	8.15	6.28	6.63	8.50	8.72	10.63	13.19	
	HDD50	6394	1442	1200	955	480	125	14	0	2	54	295	681	1146	
	HDD65	10614	1907	1620	1418	923	482	198	97	152	346	730	1130	1611	
	CDD50	1419	0	0	2	7	115	305	432	356	171	30	1	0	
	CDD65	164	0	0	0	0	7	39	64	41	13	0	0	0	
	CDH74	1657	0	0	1	4	123	455	611	350	111	2	0	0	
	CDH80	364	0	0	0	0	28	106	145	67	18	0	0	0	
Wind	WSAvg	7.3	7.0	7.3	7.7	8.0	7.7	6.9	6.6	6.4	6.9	7.6	7.8	7.2	
Precipitation	PrecAvg	34.7	2.3	2.0	1.9	2.2	2.8	3.5	4.1	3.8	3.9	3.2	3.0	2.7	
	PrecMax	40.5	4.3	3.7	3.3	3.9	5.1	5.6	8.3	5.4	5.8	4.7	5.2	4.7	
	PrecMin	31.1	1.2	1.0	0.4	0.9	1.2	1.5	2.0	1.9	2.0	1.6	2.0	1.4	
	PrecStd	2.4	0.7	0.7	0.8	0.8	1.0	1.1	1.7	1.1	1.1	0.9	0.7	0.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.6	42.2	56.6	69.9	83.8	87.6	88.9	85.8	82.3	71.6	56.6	42.0	
		MCWB	39.9	38.3	46.8	54.7	64.0	69.1	71.3	69.0	67.1	60.5	51.4	39.8	
	2%	DB	34.2	35.8	48.0	61.2	77.0	83.0	84.0	81.4	77.0	65.3	49.9	36.9	
		MCWB	33.2	32.9	40.6	47.4	59.3	66.5	68.2	67.3	65.8	56.8	46.1	35.3	
	5%	DB	30.6	31.9	42.8	55.5	72.5	79.4	80.3	77.9	72.8	60.1	45.6	34.0	
		MCWB	29.5	30.1	37.1	44.7	56.9	63.7	66.3	65.5	63.1	54.4	42.7	32.8	
	10%	DB	26.5	27.8	38.2	50.4	67.6	75.5	77.1	74.3	68.7	55.6	41.3	31.9	
		MCWB	25.3	25.8	33.4	41.9	54.7	62.0	64.6	63.8	61.5	51.0	38.6	31.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.2	39.7	50.9	56.8	68.3	72.3	73.8	71.8	70.1	62.5	51.7	40.6
			MCDB	40.7	41.8	53.6	66.9	77.4	84.1	85.0	81.4	76.7	68.1	54.9	41.5
2%		WB	33.3	33.2	42.0	49.8	62.9	69.1	70.7	69.5	67.5	58.2	46.9	35.4	
		MCDB	33.9	35.7	46.7	56.9	72.6	79.0	79.9	77.9	74.3	63.7	49.0	36.6	
5%		WB	29.5	29.9	37.4	45.8	59.5	66.5	68.6	67.2	65.0	55.0	42.9	33.0	
		MCDB	30.5	31.9	42.1	53.9	67.9	75.1	76.6	74.9	71.3	59.6	45.4	34.0	
10%		WB	25.1	25.8	33.6	42.3	56.0	64.2	66.6	65.3	62.0	51.3	38.8	30.9	
		MCDB	26.4	27.7	37.7	49.7	65.7	71.8	74.3	72.3	68.3	55.5	41.0	31.9	
Mean Daily Temperature Range	5% DB	MDBR	19.4	21.6	21.6	20.1	22.7	22.6	21.0	20.2	19.5	14.9	12.7	15.6	
		MCDBR	16.6	18.8	23.2	28.8	30.2	27.8	25.2	24.2	24.1	22.6	17.2	12.0	
		MCWBR	16.2	16.8	16.8	18.2	15.7	13.7	12.5	12.5	14.1	15.6	14.6	11.7	
	5% WB	MCDBR	16.6	18.6	20.9	25.5	24.9	22.2	20.9	20.5	20.5	20.4	15.9	11.9	
		MCWBR	16.5	17.0	15.9	17.6	15.0	12.9	11.4	11.6	13.1	15.4	14.4	11.8	
Clear-Sky Solar Irradiance	taub	0.262	0.279	0.313	0.352	0.362	0.383	0.392	0.383	0.366	0.325	0.300	0.262		
	taud	2.262	2.219	2.203	2.305	2.398	2.355	2.361	2.383	2.409	2.511	2.445	2.325		
	Ebn at Noon	257	278	285	285	285	279	274	272	266	262	243	243		
	Edh at Noon	23	31	37	38	36	38	37	35	31	24	20	20		
All-Sky Solar Radiation	RadAvg	391	735	1187	1466	1652	1799	1725	1448	1086	638	331	266		
	RadStd	61	106	122	122	146	115	146	98	102	88	38	34		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page